

BE 202: Biochemistry and Molecular Biology

Course Code	BE 202
Course Name	Biochemistry and Molecular Biology
L-T-P-C	3-0-2-4
Prerequisites	None
Intended for	B. Tech M.Tech Integrated Dual Degree Bioengineering students
Distribution	Core for Integrated Dual Degree Bioengineering students, elective for other B.Tech students
Approval	recommended by the 35th BoA (Aug 2020)
Revised	40th BoA (May 2021) and 31st Senate (Jun 2021) - renamed from "Biology II"; 49th BoA (Mar 2023) - credits changed to 3-0-2-4

Preamble

The objective of the course is to provide students with a comprehensive and concise overview of biological science with emphases on its relationship with biomedical engineering.

Course Outline

Theory: [28 hours]

- **Module 1: Cell composition:** Chemical bonds, water, pH and buffers; Amino acids, peptides and proteins; Carbohydrates: Monosaccharides, Disaccharides, Polysaccharides and glycoconjugates; Nucleotides and Nucleic acids; Lipids: Storage lipids, structural lipids, lipid as signal, cofactors and pigments [6 hours]
- **Module 2: Bioenergetics and Biochemical Reaction:** Bioenergetics and Thermodynamics, ATP, Biological oxidation-reduction reactions, Overview of central and secondary metabolism; Amino acid Oxidation: Metabolic fates of amino acids, Nitrogen excretion and urea cycle, amino acid degradation; Oxidative Phosphorylation: Electron transport reaction in mitochondria, ATP synthesis, Photophosphorylation, Light absorption and light driven electron flow, ATP synthesis by photophosphorylation [6 hours]
- **Module 3: Cellular energy:** Breakdown of Sugar: Glycolysis, Gluconeogenesis and the Pentose Phosphate Pathway; The Citric acid cycle; Fatty acid metabolism: Digestion, metabolism and transport of fats, oxidation of fatty acids, ketone bodies [8 hours]
- **Module 4: Central Dogma of Life:** Genes and Chromosomes; DNA metabolism: DNA replication, DNA repair and recombination; RNA metabolism: Transcription, RNA processing, RNA dependent RNA and DNA synthesis; Protein metabolism: The genetic code and Translation Gene Regulation [8 hours]

Lab [28 hours]

1. Preparation of buffers,
2. Quantitative determination of proteins,
3. Quantitative determination of carbohydrates and lipids,
4. Isolation of nucleic acid (DNA or RNA) and its estimation,
5. Gel Electrophoresis- Protein and nucleic acid
6. DNA amplification using PCR,
7. Isolation of chloroplasts, its estimation and assay for Hill reaction

Text Books

1. David L. Nelson, Michael M. Cox, *Lehinger Principles of Biochemistry*, 7th Edition, McMillan Learning, 2017.

References

1. Harvey Lodish, Arnold Berk, Chris A. Kaiser, Monty Krieger, Anthony Bretscher, Hidde Ploegh, Angelika Amon, Kelsey C. Martin, *Molecular Cell Biology* - 8th Edition W.H. Freeman, 2016.
2. George Plopper, David Sharp, Eric Sikorski, *Lewin's Cells* – 3rd Edition – Johns & Bartlett Publishers, 2015
3. Bruce Alberts, Alexander Johnson, Julian Lewis, Martin Raff, Keith Roberts, Peter Walter, *Molecular Biology of the Cell* - 6th Edition; Garland Science, 2014.